

Nexrutine® inhibits tumorigenesis in mouse skin and induces apoptotic cell death in human squamous carcinoma A431 and human melanoma A375 cells

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Nexrutine® (NX), a herbal extract from *Phellodendron amurense*, has been shown to possess antitumor, antimicrobial, anti-inflammatory and other biological activities. In the present investigation, we explored the mechanism of chemopreventive/chemotherapeutic efficacy of NX against skin cancer. Single application of NX (1.0 mg/mouse) prior to 12-*O*-tetradecanoylphorbol 13-acetate (TPA) application significantly inhibited TPA-induced skin edema, hyperplasia, thymidine incorporation and ornithine decarboxylase (ODC) activity; expression of cyclooxygenase-2 (COX-2) and inducible nitric oxide synthase (iNOS); phosphorylation of extracellular signal-regulated kinases (ERK) 1/2, p38 and c-jun N-terminal kinase (JNK) mitogen-activated protein kinases (MAPKs); and activation of I kappa B kinase (IKK), IκBα and nuclear factor-kappa B (NF-κB) in mouse skin. In a two-stage mouse skin tumorigenesis model, it was found that twice-weekly treatment of NX prior to TPA application in 7,12-dimethylbenz[α]anthracene (DMBA)-initiated animals showed reduced tumor incidence, lower tumor body burden and significant delay in latency period compared with DMBA-initiated and TPA-promoted animals. Furthermore, the therapeutic efficacy of NX was assessed against human squamous carcinoma (A431) and human melanoma (A375) cells. A431 and A375 cells treated with NX (2.5–10.0 μg/ml, 48 h) showed a decrease in viability and enhanced cell cycle arrest at the G₀/G₁ phase and apoptosis; however, NX had minimal cytotoxic effect on HaCaT cells and primary murine keratinocytes, suggesting its high therapeutic index. In addition, NX treatment also modulates the levels of Bax and Bcl-2 proteins along with cytochrome *c* release, cleavage and enhanced expression of poly (adenosine diphosphate-ribose) polymerase as well as catalytic activities of caspases 3 and 9 in both A431 and A375 cells. Based on our *in vivo* and *in vitro* studies, NX could be useful in the management (chemoprevention as well as chemotherapy) of skin cancer.

Introduction

Skin cancer is the most common form of cancer and it has been estimated that almost 132 000 cases of malignant melanoma (66 000 deaths) and >2–3 million cases of other skin cancers are being diagnosed annually (1). There are three main types of skin cancers: basal cell carcinoma, squamous cell carcinoma and malignant

Abbreviations: COX-2, cyclooxygenase-2; CREB, cAMP response element-binding protein; DMBA, 7,12-dimethylbenz[α]anthracene; ERK, extracellular signal-regulated kinase; HRP, horseradish peroxidase; IKK, I kappa B kinase; iNOS, inducible nitric oxide synthase; JNK, c-jun N-terminal kinase; MAPK, mitogen-activated protein kinase; MTT, 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide; NF- B, nuclear factor-kappa B; NX, nexrutine; ODC, ornithine decarboxylase; PARP, poly (adenosine diphosphate-ribose) polymerase; PMK, primary murine keratinocyte; TPA, 12-*O*-tetradecanoylphorbol 13-acetate.

melanoma. Among these, malignant melanoma is the most aggressive form of skin cancer as it has the ability to metastasize, resulting in the death of 75% of skin cancer patients. Diagnosis of skin cancer at an early stage can be favorably cured by surgical removal of tumor tissue; however, the response of single-agent chemotherapy in patients with metastatic disease is <15% (2). Therefore, it is necessary to develop novel approaches for prevention and treatment of skin cancer.

In recent years, considerable efforts have been made to search for naturally occurring substances that can be used in the intervention of carcinogenesis (3,4). One such agent is Nexrutine® (NX), a commercially available herbal extract from *Phellodendron amurense*, which is widely used for the treatment of inflammation, gastroenteritis, abdominal pain and diarrhea and is shown to exhibit minimal toxicity to normal tissues (5). NX is reported to contain isoquinoline alkaloids, phenolic compounds and flavone glycosides (6). Recently, NX has been demonstrated to inhibit proliferation of prostate and lung cancer cells through the modulation of the protein kinase B (Akt)- and cAMP response element-binding protein (CREB)-mediated signaling pathways and its antiproliferative effects are comparable with that of berberine, which is a well-known chemopreventive agent (7–9). In addition, it has been noted that NX prevents early stages of tumor development in prostate cancer as well as progression of tumors in a transgenic adenocarcinoma of mouse prostate (TRAMP) model (5,10). Thus, it may be inferred that NX is a potent anticancer agent at least for prostate and lung cancer; however, its anticarcinogenic activity in other tissues has not been evaluated.

In the present study, skin cancer chemopreventive activity of NX was evaluated by assessing its anti-inflammatory and anti-tumor-promoting potential in a multistage mouse skin carcinogenesis model. In addition, for evaluation of the chemotherapeutic potential of NX against skin cancer, human squamous A431 cells and human melanoma A375 cells were exposed to NX to determine its efficacy as a growth inhibitory agent and elucidate the mechanisms underlying its antitumor activity.

Materials and methods

Preparation of Nexrutine

NX was provided by Next Pharmaceuticals (Irvine, CA). Stock solution of NX was prepared by dissolving 10.0 mg of NX in 10.0 ml dimethyl sulfoxide to obtain a concentration of 1 mg/ml. The stock solution was further diluted in growth medium to obtain various working concentrations.

Antibodies and chemicals

Antibodies specific for cyclooxygenase-2 (COX-2) and inducible NO synthase (iNOS) were procured from Cayman Chemical Company (Ann Arbor, MI). Antibodies specific for phosphorylated extracellular signal-regulated kinases (p-ERK)1/2, ERK1/2, p38, c-jun N-terminal kinase (JNK), I kappa B kinase (IKK), IκBα, cytochrome *c*, cleaved poly (adenosine diphosphate-ribose) polymerase (PARP), α-tubulin and β-actin-horse radish peroxidase (HRP) were purchased from SantaCruz Biotechnology (Santa Cruz, CA), whereas antibodies specific for p-p38, p-JNK, p-NF-κB, p-IκBα, p-IKK, Bax, Bcl-2, cleaved caspase 3 and cleaved caspase 9 were purchased from Cell Signaling (Beverly, MA). Dulbecco's modified Eagle's medium, RPMI-160, streptomycin, penicillin, fetal bovine serum, trypsin/ethylenediamine tetraacetic acid solution, 3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyl tetrazolium bromide (MTT), dithiothreitol, phenylmethylsulfonyl fluoride, 2-β-mercaptoethanol, propidium iodide, RNase A, protease inhibitor cocktail set-I, ethidium bromide, ethylenediamine tetraacetic acid disodium salt, Tris buffer, Triton X-100, Tween-20, Tween-80 and ornithine were purchased from Sigma Chemicals Co. (St. Louis, MO). [³H]-thymidine and D,L-[¹⁴C]-ornithine were purchased from Amersham (Arlington Heights, IL). All other chemicals and reagents used were of highest purity commercially available.

Animals

Six- to seven-week-old female Swiss albino mice (20 ± 3 g), derived from the animal breeding colony of CSIR-Indian Institute of Toxicology Research (CSIR-IITR), Lucknow, were acclimatized under standard laboratory conditions and given a commercial pellet diet (Ashirwad Industries, Chandigarh, India) and water *ad libitum*. Animals were housed in plastic cages having rice husk as bedding and maintained in controlled atmosphere of 12 h dark/light cycle, $22 \pm 2^\circ\text{C}$ temperature and 50–60% humidity as per rules laid down by the Animal Welfare Committee of CSIR-IITR. The mice were shaved with an electric clipper (Oster, WI) 1 week prior to the beginning of the experiment. Mice showing no signs of hair growth were used for further experiments. All the experiments involving animals were approved by the Institutional Animal Ethics Committee, CSIR-IITR, Lucknow. Animals were killed by cervical dislocation with minimal suffering as per CSIR-IITR guidelines.

Edema

To assess the inhibitory effect of preapplication of NX on TPA-induced skin edema, the animals were divided into eight groups having five mice per group. The mice of the first group received a single topical application of 0.2 ml acetone, and animals of second, third and fourth groups received a single topical application of 0.33, 1.0 or 3.0 mg NX/mouse, respectively. The mice of the fifth group received a single topical application of TPA (4.0 nmol/mouse), whereas animals of the sixth, seventh and eighth groups received a single topical application of 0.33, 1.0 and 3.0 mg NX/mouse 30 min prior to a single topical application of TPA (4.0 nmol/mouse). In our studies, we chose the doses of 0.33, 1.0 and 3.0 mg NX based on the previous reported dose of NX administered through diet (600 mg/kg diet) (5), which is equivalent to 3.0 mg NX/mouse considering 5.0 gm of diet consumption/day/mouse (11). The animals were killed at 6, 24 and 48 h after TPA application and 1-cm-diameter punches of skin from control-, NX-, TPA- or NX-plus-TPA-treated animals were removed, made free of fat and weighed. After drying for 24 h at 50°C , the skin punches were weighed again and the loss of water content was determined. The difference in the amount of water gain between the control (vehicle-treated) and TPA-treated skins represented the extent of edema induced by TPA, whereas the difference between the TPA and NX-plus-TPA groups represented the inhibitory effect of NX on TPA-induced edema.

Thymidine incorporation and ornithine decarboxylase enzyme activity in cutaneous tissue

To assess the inhibitory effect of preapplication of NX on TPA-induced thymidine incorporation, the animals were divided into four groups having five mice per group. The mice of the first group received a single topical application of 0.2 ml acetone, and animals of the second group received a single topical application of 1.0 mg NX/mouse. The mice of the third group received a single topical application of TPA (4.0 nmol/mouse), whereas animals of the fourth group received a single topical application of 1.0 mg NX/mouse 30 min prior to a single topical application of TPA (4.0 nmol/mouse). Animals of all the four groups were given intraperitoneal injection of [^3H]-thymidine (30.0 μCi /animal in 0.2 ml normal saline) 2 h prior to the termination of the experiment. After 20 h of treatment, animals were killed and skin was excised. Epidermal DNA was isolated and assessment of incorporation of [^3H]-thymidine into DNA was carried out according to the method of Gupta and Mehrotra (12).

For estimation of ornithine decarboxylase (ODC) activity, animals were treated as described above. After 4 h of treatment, skin was dissected out and ODC activity was measured by the method of Verma et al. (13). The specific activity was expressed as pmol $^{14}\text{CO}_2$ released/h/mg protein.

Immunohistochemistry of COX-2 and iNOS

To assess the inhibitory effect of preapplication of NX on TPA-induced expression of COX-2 and iNOS proteins, the animals were divided into eight groups having three mice per group. The mice of the first group received a single topical application of 0.2 ml acetone, and animals of the second group received a single topical application of 1.0 mg NX/mouse. The mice of the third, fifth and seventh groups received a single topical application of TPA (4.0 nmol/mouse), whereas animals of the fourth, sixth and eighth groups received a single topical application of 1.0 mg NX/mouse 30 min prior to a single topical application of TPA (4.0 nmol/mouse). The animals were killed at 6, 24 and 48 h after TPA application and the excised skin was fixed in 10% neutralized formalin. Immunohistochemistry for COX-2 and iNOS was performed in skin sections using Super Sensitive Polymer-HRP Detection System from BioGenex (San Ramon, CA) as per the manufacturer's instructions.

Preparation of cell extracts and western blot analysis

For the *in vivo* western blots, animals were divided into eight groups having three mice per group. The animals were treated as described in the section "Immunohistochemistry for COX-2 and iNOS" and killed by cervical dislocation at 6, 24 and 48 h. Skin was excised, made free of fat and homogenized in ice-cold lysis buffer supplemented with protease and phosphatase inhibitors; the cytoplasmic and nuclear extracts were prepared as described earlier (14).

For the *in vitro* western blots, A431 and A375 cells were treated with or without NX (0, 2.5, 5.0 and 10.0 $\mu\text{g}/\text{ml}$) and after 48 h, cells were harvested, washed with cold phosphate-buffered saline, and lysed with ice-cold RIPA (Radio-immunoprecipitation Assay) buffer supplemented with protease inhibitors. Proteins (40 μg) were subjected to 10% sodium dodecyl sulfate–polyacrylamide gel electrophoresis, transferred to a polyvinylidene difluoride membrane (Millipore, Billerica, MA) and incubated with specific primary antibodies at 4°C overnight, followed by incubation with HRP-conjugated secondary antibody (Sigma, St. Louis, MO). Bound antibody was detected by enhanced chemiluminescence using Amersham ECL Western Blotting Detection Reagents, following the manufacturer's instructions (Amersham, Fairfield, CT). All the blots were stripped and reprobed with β -actin or α -tubulin to ensure equal loading of protein.

Skin tumorigenesis

Female Swiss albino mice were used for 7,12-dimethylbenz[α]anthracene (DMBA)-initiated and TPA-promoted two-stage skin tumorigenesis protocol. The dorsal side of the skin was shaved using an electric clipper, and mice with hair cycles in the resting phase were used for tumor studies. The animals were randomly divided into three groups having 10 mice per group. Tumor induction was initiated by a single topical application of 120.0 nmol DMBA in 0.2 ml acetone, and 1 week later, these animals were given twice-weekly topical applications of 4.0 nmol TPA in 0.2 ml acetone as promoter. Treatment with TPA alone or with NX (1.0 mg/mouse) 30 min prior to TPA application was given twice every week up to the termination of the experiment at 24 weeks. Animals in both the groups were observed for any apparent signs of toxicity, such as weight loss or mortality, during the entire period of study. Skin tumor formation was recorded weekly, and tumors >1 mm in diameter were included in the cumulative number if they persisted for 2 weeks or more.

Cell culture

Human squamous carcinoma cells (A431) and human epithelial malignant melanoma cells (A375) were obtained from the National Centre for Cell Science, Pune, India. Human keratinocytes (HaCaT) were a kind gift from Prof. Susan M. Fischer, The University of Texas M.D. Anderson Cancer Center, Science Park, Smithville, TX. A431 and HaCaT cells were cultured in Dulbecco's modified Eagle's medium, whereas A375 cells were grown in RPMI-1640 supplemented with 10% fetal bovine serum and 1% penicillin–streptomycin solution. Primary murine keratinocytes (PMKs) from newborn Swiss albino mice were prepared as described earlier (15). The cells were maintained under standard cell culture conditions at 37°C and 5% CO_2 in a humid environment.

Cell viability assay

The effect of NX on the viability of cells was determined by the MTT assay as described earlier (16). The effect of NX on cell viability was assessed as the percentage of cell viability compared with vehicle-treated control cells, which were arbitrarily assigned 100% viability.

Analysis of apoptotic cell death and cell cycle distribution by flow cytometry

A431 and A375 cells (60–70% confluent) were treated with NX (2.5, 5.0 and 10.0 $\mu\text{g}/\text{ml}$) for 48 h in complete medium. After incubation, both adherent and floating cells were collected by trypsinization, washed twice with ice-cold phosphate-buffered saline and 5.0×10^5 cells were analyzed for apoptosis using Annexin V–fluorescein isothiocyanate Apoptosis Detection Kit (BD Pharmingen, San Jose, CA) as per the manufacturer's instructions with a FACS Canto II (Becton Dickinson, Franklin Lakes, NJ) flow cytometer. For cell cycle analysis, 5.0×10^5 cells were fixed in 70% ethanol for 1 h at -20°C and subsequently incubated with propidium iodide (20.0 $\mu\text{g}/\text{ml}$) and RNase A (200.0 $\mu\text{g}/\text{ml}$) for another 30 min at 37°C . A minimum of 10 000 events per sample were acquired and the DNA histograms were analyzed by FACSDiva software (Becton Dickinson, Franklin Lakes, NJ).

Caspase activity assay

Following treatment of the A431 and A375 cells with or without NX, the cells were harvested, washed with ice-cold phosphate-buffered saline, and lysed with ice-cold lysis buffer [10.0 mM Tris (pH 7.5), 10.0 mM NaH_2PO_4 , 130.0 mM NaCl, 1% Triton X-100, 10.0 mM sodium pyrophosphate, 1.0 mM dithiothreitol and 250.0 μM phenylmethylsulfonyl fluoride] by incubating in

ice for 30 min with intermittent vortexing. After centrifugation, the supernatant was collected, quantified and 100.0 µg of protein from each sample was subjected to respective caspase activity assay using specific substrates conjugated with 7-amino-4-trifluoromethylcoumarin as fluorophore (BD Pharmingen, San Jose, CA and Biovision Inc, CA), followed by incubation at 37°C for 1 h. The fluorescence intensity was measured at excitation and emission wavelengths of 400 and 505 nm, respectively, using a microplate reader (Synergy HT; Biotek, VT).

Statistical analysis

The results were expressed as the mean ± standard error (SE). The statistical significance of difference between the values of control and treatment groups was determined using either two-tailed Student's *t*-test or chi-square test. A *P* value of <0.05 was considered statistically significant.

Results

Inhibitory effect of NX on TPA-induced cutaneous edema, thymidine incorporation and ODC activity

Inhibitory effect of topical application of NX against TPA-induced cutaneous edema was assessed. The preapplication of NX showed significant protection (30–39%) against TPA-induced edema measured at 6, 24 and 48 h. NX at the dose of 0.33 mg/mouse showed no significant effect on TPA-induced cutaneous edema (Figure 1A). Because NX exhibited significant inhibitory effect on TPA-induced edema at both the doses of 1.0 and 3.0 mg/mouse, we selected the dose of 1.0 mg NX/mouse for further studies.

Figure 1B shows the effect of preapplication of NX on TPA-induced cell proliferation. Topical application of NX (1.0 mg/mouse) resulted in significant inhibition (41%) of TPA-induced [³H]-thymidine incorporation in mouse skin. In order to determine the effect of NX against TPA-induced ODC activity, animals were treated topically with NX (1.0 mg/mouse) prior to TPA application (4.0 nmol/mouse). As shown in Figure 1C, preapplication of NX resulted in significant inhibition (69%) of TPA-induced epidermal ODC activity.

Inhibitory effect of NX on TPA-induced epidermal COX-2 and iNOS protein expression

COX-2 and iNOS are well-established molecular biomarkers of inflammation and tumor promotion and could be promising molecular targets for the design of drugs targeting cancer prevention as well as therapy (4). In the present study, we observed that COX-2 and iNOS protein expression was elevated in mouse skin treated with TPA for 6, 24 and 48 h, respectively (Figure 2A). Subsequently, topical application of NX (1.0 mg/mouse) prior to TPA treatment (4.0 nmol/mouse) resulted in substantial decrease in COX-2 and iNOS protein levels in mouse skin, as observed by western blot analysis. These results were further validated by immunohistochemistry (Figure 2B and 2C), where substantial increase in protein expressions of COX-2 and iNOS in the epidermis of TPA-treated animals was observed. In contrast, topical application of NX (1.0 mg/mouse) prior to TPA application (4.0 nmol/mouse) showed a significant decrease in the expression levels of both the proteins at the 6, 24 and 48 h time points. Further, results in Figure 2B and 2C also showed a decrease in hyperplasia following NX exposure to TPA-treated animals. These results suggest that NX has the ability to suppress TPA-induced inflammation by downregulating COX-2 and iNOS in the mouse skin.

Inhibitory effect of NX on TPA-mediated phosphorylation of mitogen-activated protein kinases

To determine the inhibitory effect of NX on TPA-induced activation of ERK1/2, p38 and JNK mitogen-activated protein kinases (MAPKs) in mouse skin, western blot analysis was performed. Topical application of TPA to mice resulted in increased phosphorylation of ERK1/2, p38 and JNK at 6, 24 and 48 h, respectively (Figure 2D). However, topical application of NX (1.0 mg/mouse) prior to TPA application (4.0 nmol/mouse) resulted in significant inhibition of TPA-induced phosphorylation of ERK1/2, p38 and JNK at these selected time points (Figure 2D).

Inhibitory effect of NX on TPA-induced activation of nuclear factor-kappa B and IKKα and phosphorylation and degradation of IκBα protein expression

Studies have shown that one of the critical events in nuclear factor-kappa B (NF-κB) activation is its dissociation from inhibitory protein IκBα via phosphorylation and ubiquitination (17). To determine whether the inhibitory effect of NX was attributable to an effect on IκBα phosphorylation, we examined the level of p-IκBα in mouse skin. TPA application to mouse skin resulted in the phosphorylation of IκBα protein at 6, 24 and 48 h after treatment. However, topical application of NX prior to TPA treatment resulted in inhibition of TPA-induced phosphorylation of IκBα protein at selected time points (Figure 2E). It was also observed that TPA application resulted in the activation of IKKα protein, which in turn phosphorylates and degrades IκBα protein. Topical application of NX prior to TPA inhibited phosphorylation of IKKα (Figure 2E). Furthermore, in the nuclear fraction, TPA application to mouse skin resulted in activation and increased nuclear translocation of NF-κB/p65; however, topical application of NX prior to TPA application inhibited NF-κB/p65 activation and nuclear translocation at selected time points (Figure 2E).

Inhibitory effect of NX on TPA-induced skin tumor promotion

As shown in our *in vivo* studies, NX significantly inhibited TPA-induced short-term markers of inflammatory responses. Therefore, it was of relevance to assess the efficacy of preapplication of NX on TPA-mediated skin tumor promotion in DMBA-initiated mice. As shown in Figure 3, although topical application of NX prior to TPA treatment in DMBA-initiated mouse skin did not result in complete abolition of tumor formation, it substantially delayed the latency period from 6 to 11 weeks and afforded protection when tumor data were considered in terms of tumor incidence and multiplicity throughout the treatment period (*P* < 0.05, chi-square test). Similarly, compared with a total of 68 tumors in the DMBA/TPA-treated group, only 15 tumors were observed in the NX-treated group (Figure 3A). In the DMBA/TPA-treated group, 100% of the mice developed tumors at 11 weeks on test, whereas in the group which was treated with NX 30 min prior to TPA application, only 11% of the mice exhibited tumors at this time point (Figure 3B). At the termination of the experiment at 24 weeks, only 50% of the mice that received topical application of NX prior to each TPA application showed tumor development. Also, compared with 6.8 tumors/mouse in the DMBA/TPA-treated group, only 3 tumors per mouse were observed in the NX-treated group at 24 weeks (Figure 3C). When these tumor data were considered in terms of number of tumors per mouse at the termination of experiment, compared with 6.8 tumors per mouse in the DMBA/TPA group, only 1.5 tumors/mouse in the NX-treated group were observed (Figure 3D).

NX treatment resulted in inhibition of cell growth in A431 and A375 cells but not in PMKs and HaCaT cells

The inhibitory effect of NX (0.5–20.0 µg/ml) on the growth of human squamous carcinoma cells A431, human malignant melanoma cells A375, immortalized human keratinocytes HaCaT and normal PMKs was assessed by the MTT assay and is shown in Figure 4A and 4B. Treatment with NX (0.5–20.0 µg/ml) for 24 h decreased the cell viability in A431 (3–44%) and A375 (5–58%) cells; however, at 48 h, the effects on cell viability were more pronounced in A431 (4–70%) and A375 (5–81%) cells. Moreover, a minimal effect on HaCaT and PMK cells was noticed following NX exposure at both the time points (Figure 4A and 4B). Based on these findings, we selected NX doses of 2.5, 5.0 and 10.0 µg/ml and the 48 h time point for further studies.

NX treatment results in G₀/G₁ phase cell cycle arrest and apoptosis in A431 and A375 cells

Figure 4C shows the effect of NX on the cell cycle phases in A431 and A375 cells. Exposure of NX (2.5–10.0 µg/ml) to A431 and A375 cells for 48 h resulted in a significant increase in the proportion of cells in the G₀/G₁ phase (65–75%), with a concomitant decrease in the S (11–21%) and G₂/M (12–18%) phases. To assess whether exposure to NX

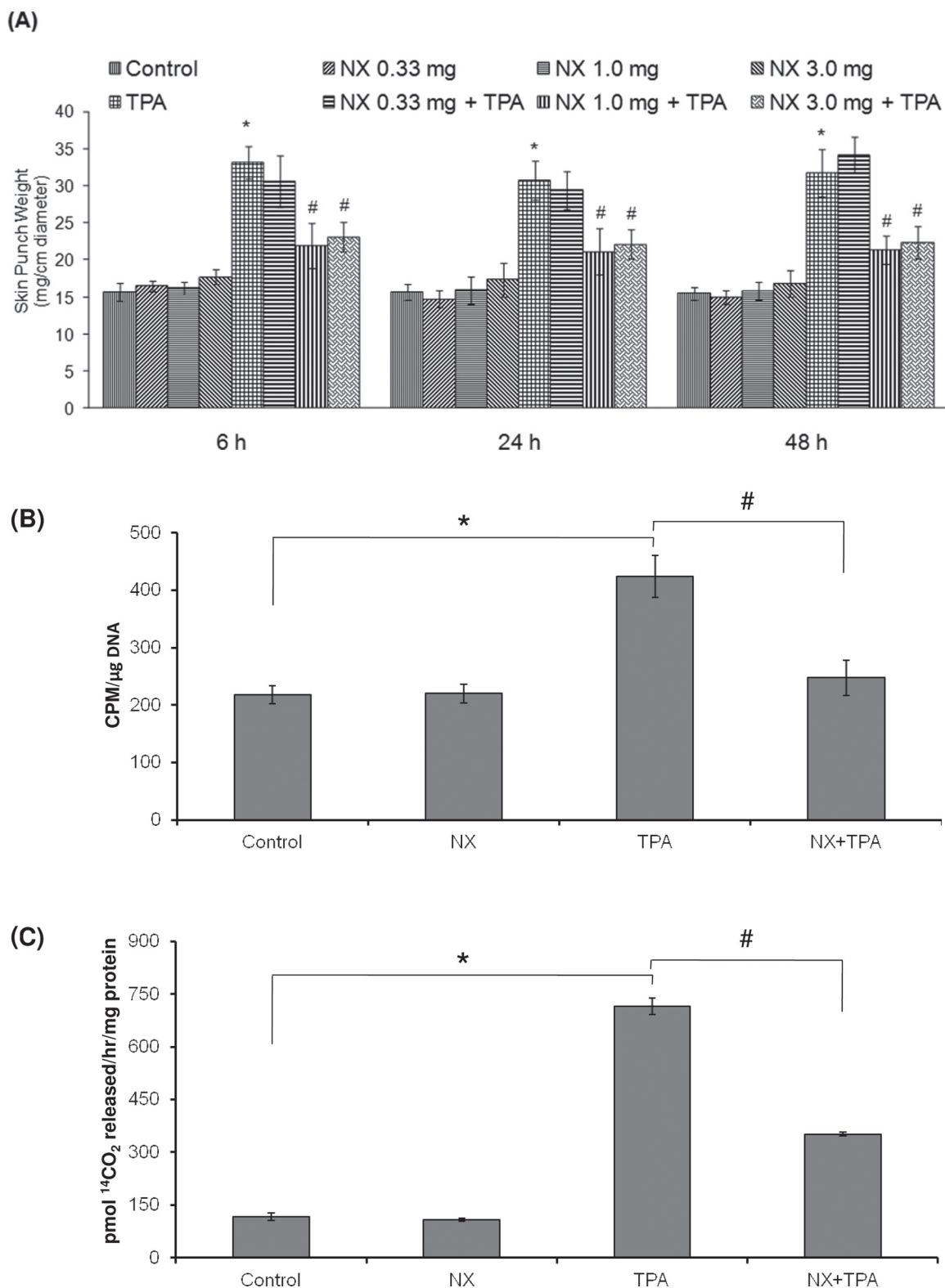


Fig. 1. Inhibitory effect of NX on TPA-induced inflammatory markers, viz., skin edema, thymidine incorporation, and ODC activity, in Swiss albino mice. **(A)** 6, 24 and 48 h after TPA or NX-plus-TPA treatment, skin edema was determined by weighing 1.0-cm-diameter punch skin as described in text. At least four determinations were made at different dorsal skin sites per mouse in each group. The data represent the mean $\pm$ standard error (SE) of five mice ($*P < 0.05$ versus control group, $#P < 0.05$ versus TPA-treated group). **(B)** The animals were killed at 20 h after TPA or NX-plus-TPA treatment, and incorporation of [³H]-thymidine in epidermal DNA was assessed as described in text. The data are shown as [³H]-thymidine incorporation (cpm/ μ g DNA) and represent the mean $\pm$ SE of five mice ($*P < 0.05$ versus control group, $#P < 0.05$ versus TPA-treated group). **(C)** The animals were killed at 4 h after TPA or NX-plus-TPA treatment, and ODC activity in mouse skin was assessed as described in text. The data are shown as ODC activity (pmol/h/mg protein) and represent the mean $\pm$ SE of five mice; each assay was performed in duplicate ($*P < 0.05$ versus control group, $#P < 0.05$ versus TPA-treated group).

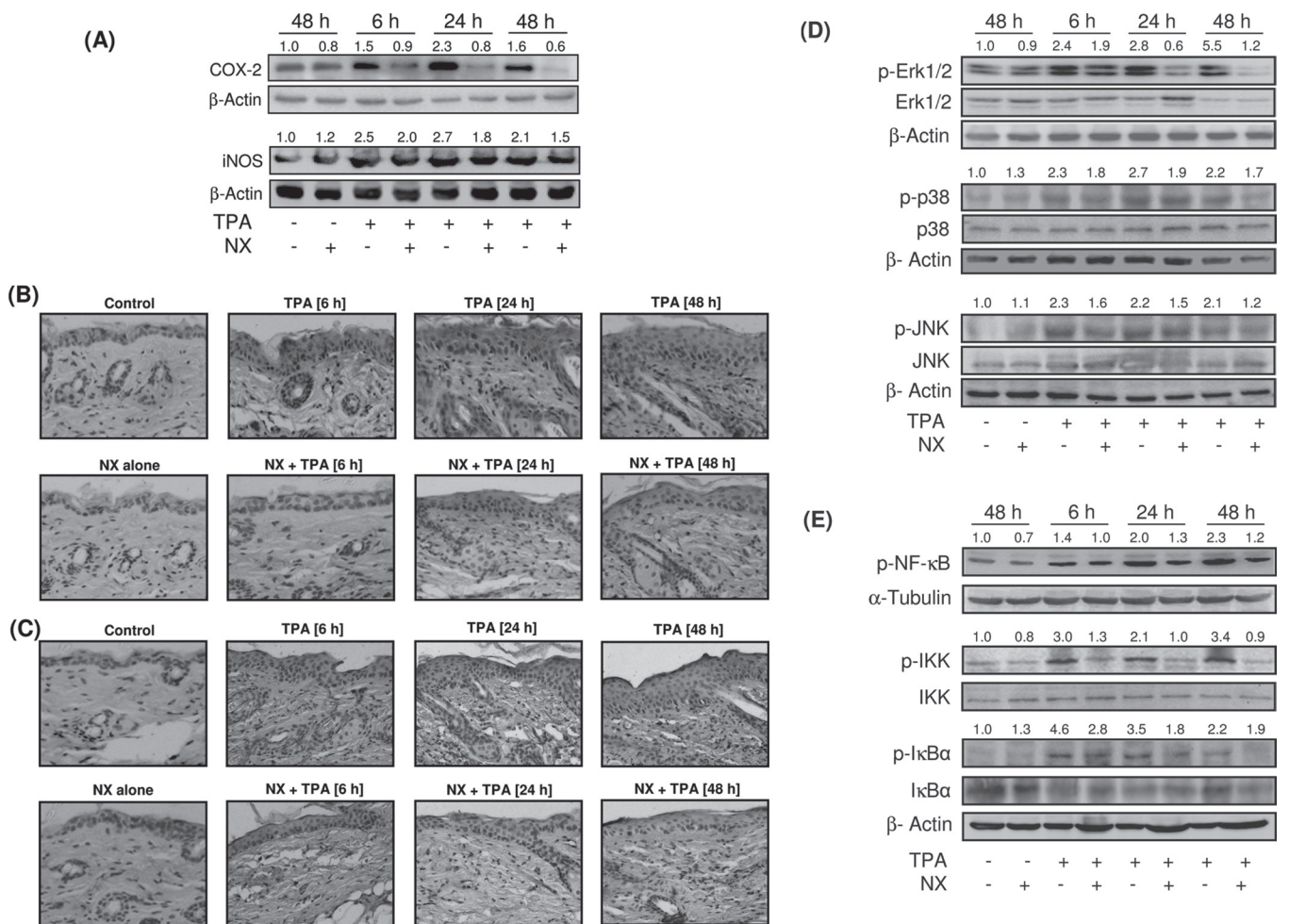


Fig. 2. Inhibitory effect of NX on TPA-induced epidermal COX-2 and iNOS protein expression, phosphorylation of MAPKs, activation of NF-κB and IKKα/β, phosphorylation and degradation of IκBα in Swiss albino mice. **(A)** At indicated time points after TPA or NX-plus-TPA treatment, the animals were killed, skin lysates were prepared and COX-2 and iNOS protein expression was determined by western blot analysis as described in text. Equal loading was confirmed by stripping the blot and reprobing it for β-actin. The blots shown here are representative of three independent experiments with similar results. Immunohistochemistry of **(B)** COX-2 and **(C)** iNOS. At different time points after TPA or NX-plus-TPA treatment, the animals were killed; the skin-punch biopsies were fixed in 10% neutralized formalin and embedded in paraffin. Five-micrometer sections were cut, deparaffinized in xylol, rehydrated with 70% ethanol, washed in phosphate-buffered saline and immunohistochemistry of COX-2 and iNOS was performed as mentioned in text. A representative picture from three independent immunohistochemistry tests is shown. **(D)** and **(E)** At different time points after TPA or NX-plus-TPA treatment, the animals were killed; the epidermal cytosolic and nuclear lysates were prepared and protein expression was determined as described in text. Equal loading was confirmed by stripping the blot and reprobing it for β-actin (in the case of cytosolic extract) and α-tubulin (in the case of nuclear extract). The blots shown here are representative of three independent experiments with similar results. The values above the figures represent relative density in terms of fold change compared with control after normalization with either total protein or β-actin.

may cause apoptosis in these cells, Annexin V–fluorescein isothiocyanate and propidium iodide dual staining was performed. Apoptotic cell death was found to be increased in NX (2.5–10.0 µg/ml)-treated A431 (4- to 7-fold) and A375 (8- to 10-fold) cells (Figure 4D).

NX treatment results in activation of intrinsic pathway of apoptosis in A375 and A431 cells

Bax and Bcl-2 proteins play a central regulatory role in apoptotic cell death. Therefore, the expression levels of Bax and Bcl-2 in both the cell lines following NX treatment were measured by western blot analysis. As shown in Figure 5A, NX treatment (2.5–10.0 µg/ml) resulted in a dose-dependent increase in the expression level of Bax and a decrease in the expression level of Bcl-2. However, the increment in Bax/Bcl-2 ratio was found to be substantially more in A375 cells compared with that in A431 cells. To further confirm

whether modulation of Bax/Bcl-2 ratio is responsible for the release of cytochrome *c* into the cytosol, A431 and A375 cells were treated with NX. As shown in Figure 5A, the levels of cytochrome *c* in the cytosol were found to be significantly elevated in a dose-dependent manner following NX treatment to both the cell lines.

It is well documented that the apoptotic process is executed by cysteinyl-aspartate-specific proteases known as caspases, which demolish the cell in an orderly fashion by cleaving a large number of cellular protein substrates (18). Therefore, activation of the caspases 3 and 9 was assessed in both the cell lines after NX treatment by western blot analysis of cleaved caspases and by measurement of catalytic activity using specific substrates. Results indicate that NX treatment to cells caused increased levels of cleaved caspases 3 and 9 as well as significant enhancement in the catalytic activities of both the caspases in a dose-dependent manner; however, the catalytic activity of

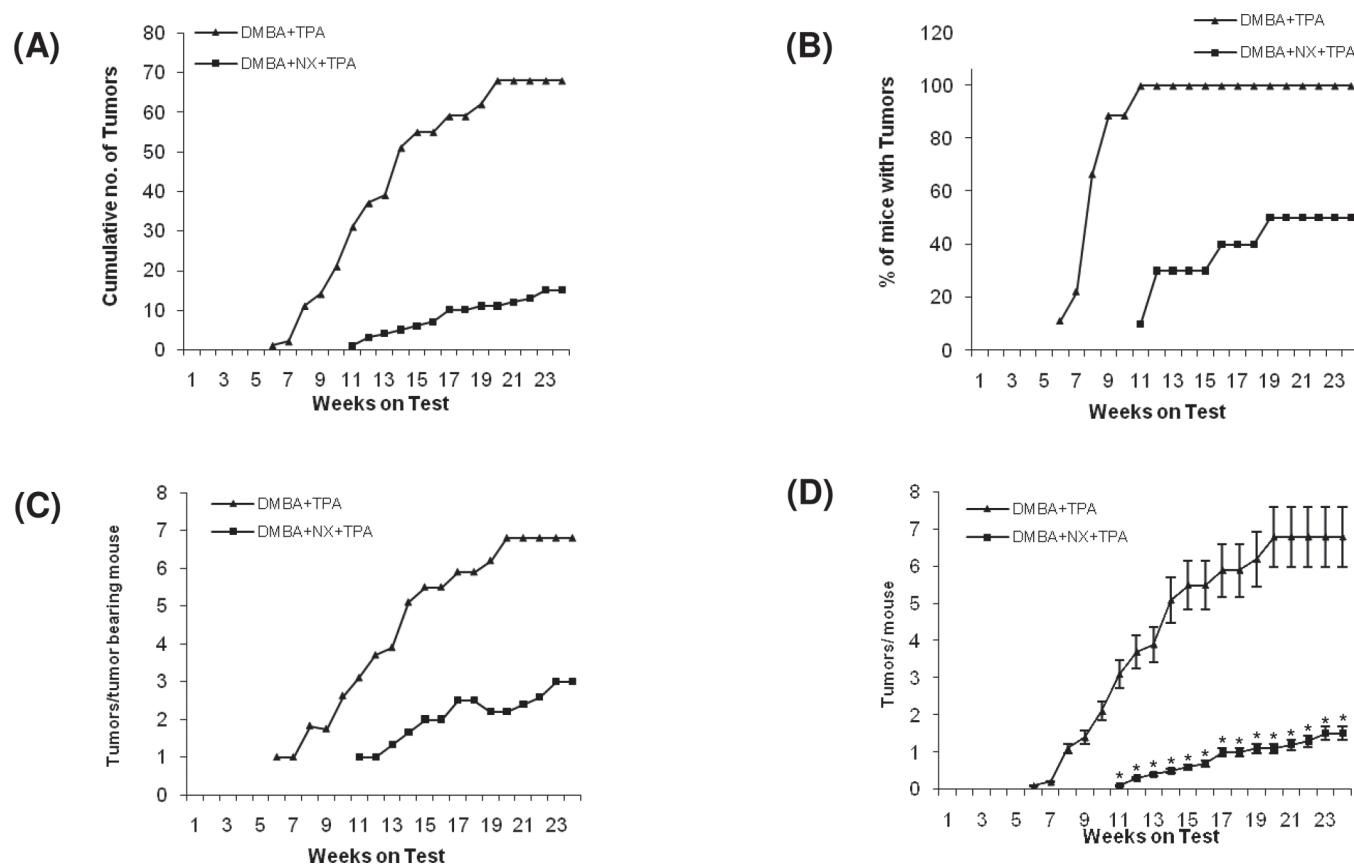


Fig. 3. Inhibitory effect of NX on DMBA-initiated and TPA-promoted tumor formation in Swiss albino mice. In each group, 10 animals were used. Tumorigenesis was initiated in the animals by a single topical application of 120.0 nmol DMBA in 0.2 ml vehicle on the dorsal shaved skin, and 1 week later, the tumor growth was promoted with twice-weekly applications of 4.0 nmol TPA in 0.2 ml vehicle. To assess its anti-skin tumor-promoting effect, NX at a dose of 1.0 mg per animal was applied topically 30 min prior to each TPA application in different groups. Treatment with TPA alone or NX-plus-TPA was repeated twice weekly up to the termination of the experiments at 24 weeks. Animals in all the groups were watched for any apparent signs of toxicity, such as weight loss or mortality, during the entire period of study. Skin tumor formation was recorded weekly, and tumors > 1 mm in diameter were included in the cumulative number only if they persisted for 2 weeks or more. The tumor data are represented as (A) the number of tumors per group, (B) the percentage of mice with tumors, (C) the number of tumors per tumor-bearing mouse and (D) the number of tumors per mouse (* $P < 0.05$ versus DMBA/TPA-treated group).

caspase 8 was not found to be altered (Figure 5B and 5C). Activation of caspase 3 results in PARP cleavage, which was also observed in a dose-dependent manner following NX treatment in both the cell lines (Figure 5A).

Discussion

The present study demonstrates that NX inhibits TPA-mediated tumor promotion in DMBA-initiated mice, which is associated with a decrease in proliferation index together with inhibition of ODC activity and protein levels of COX-2 and iNOS. The inhibition of COX-2 and iNOS levels by NX may be due to the inhibition of the NF- κ B and MAPKs pathways. Apart from its anti-tumor-promoting activity, we also observed that NX causes apoptotic cell death to human squamous carcinoma A431 and human melanoma A375 cells without affecting normal HaCaT and PMK cells, suggesting its high therapeutic index. Taken together, these *in vivo* and *in vitro* studies provide strong evidence that NX could be useful in the management (chemoprevention as well as chemotherapy) of skin cancer.

TPA is primarily a cell-proliferating agent, which leads to enhancement in edema, hyperplasia and cell proliferation that results in inflammation and ultimately skin tumor promotion (19,20). Because TPA-induced inflammation and tumor promotion in mouse skin has

been closely associated with induction of ODC, COX-2 and iNOS (21,22), inhibition of these enzymes during carcinogenesis has been recognized as an important and commonly accepted approach to effectively prevent tumor promotion (3,23,24). In the present study, topical application of NX prior to TPA application in mice resulted in significant inhibition of TPA-induced edema, hyperplasia and [3 H]-thymidine incorporation, as well as epidermal ODC activity and the expression of COX-2 and iNOS, suggesting the *in vivo* anti-inflammatory and antiproliferative potential of NX. In agreement with the above findings, earlier studies have shown that NX is responsible for inhibition of prostate and lung cancer cell proliferation through inhibition of Akt/CREB pathway-mediated cyclin D1 and COX-2 expression (5,7,9,25).

The expression of COX-2 and iNOS is reported to be regulated by NF- κ B transcription factor in various cell lines as well as in TPA-induced cutaneous inflammation (26,27). Earlier studies have shown that NF- κ B plays a crucial role in skin carcinogenesis and it is implicated in cell proliferation, apoptosis, adhesion and inflammatory responses (28,29). Recently, NX and its butanol extract fraction have been shown to inhibit NF- κ B transcriptional activity and thus decrease proliferation in prostate cancer cells (8). Our data showed that topical application of NX prior to TPA application on mouse skin inhibited IKK α activation and phosphorylation of I κ B α protein,

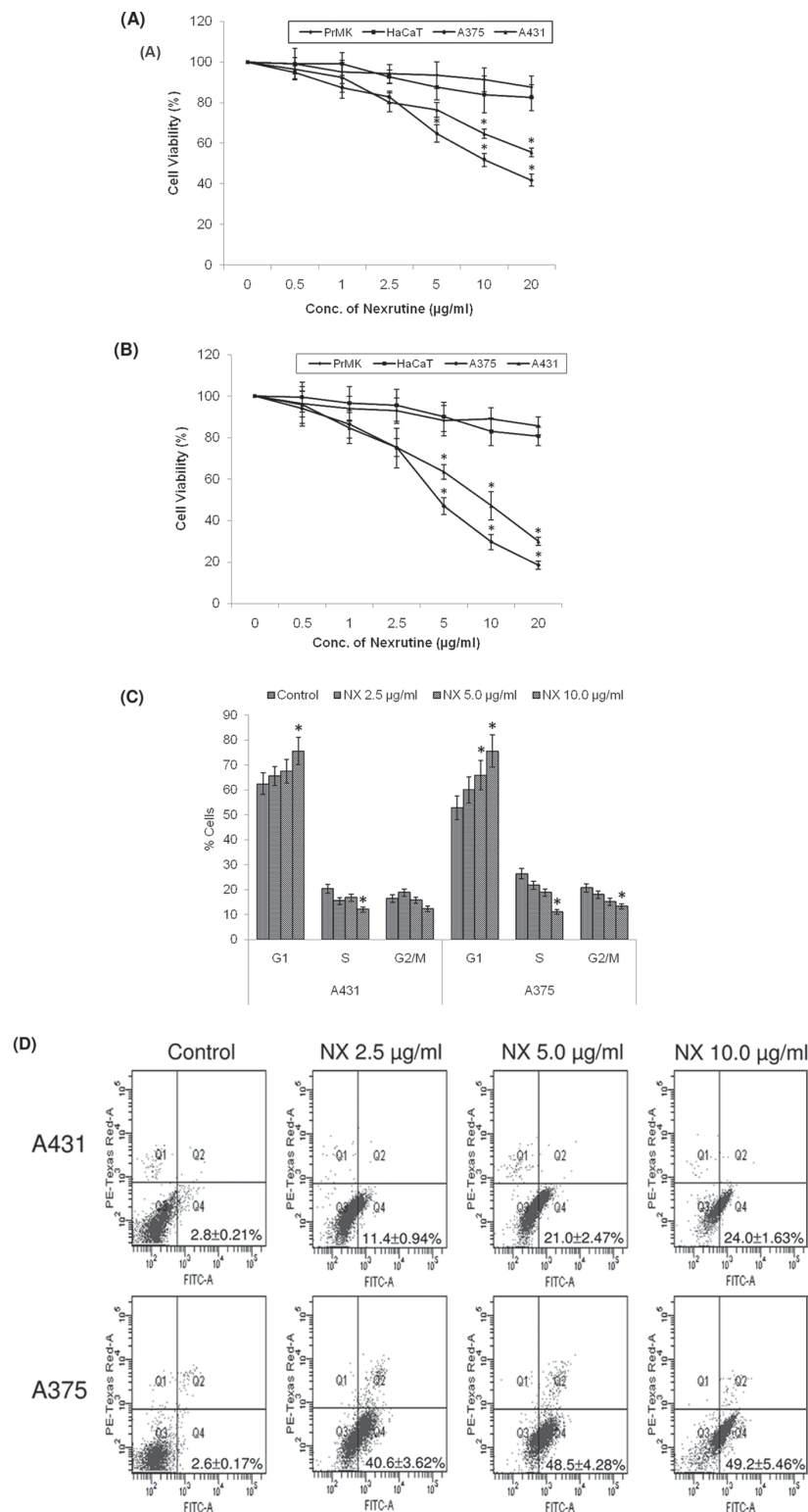


Fig. 4. (A and B) Effect of NX on cell growth. As detailed in text, PrMK, HaCaT, A375 and A431 cells were treated with NX (0.5–20.0 μg/ml) for 24 and 48 h and the viability of cells was determined by the 3-[4,5-dimethylthiazol-2-yl]-2,5-diphenyl tetrazolium bromide (MTT) assay. The data are shown as the percentage of cell viability and represent the means $\pm$ SE of five experiments in which each treatment was performed in multiple wells. (C) Effect of NX on cell cycle phase distribution in A431 and A375 cells. The cells were treated with NX (2.5–10.0 μg/ml) for 48 h and collected, stained with propidium iodide solution and data were acquired by flow cytometry as described in text. The data are shown as percentage of cells and represent the mean $\pm$ SE of five experiments in which each treatment was performed in multiple flasks (* P < 0.05 versus control). (D) Effect of NX on induction of apoptosis in A431 and A375 cells as assessed by Annexin-V–fluorescein isothiocyanate staining. The cells were treated with NX (2.5–10.0 μg/ml) for 48 h and collected, stained with Annexin-V–fluorescein isothiocyanate and propidium iodide and data were acquired by flow cytometry as described in text. The data are shown as percentage of cells in the Q2 quadrant and represent the mean $\pm$ SE of five experiments in which each treatment was performed in multiple flasks (* P < 0.05 versus control).

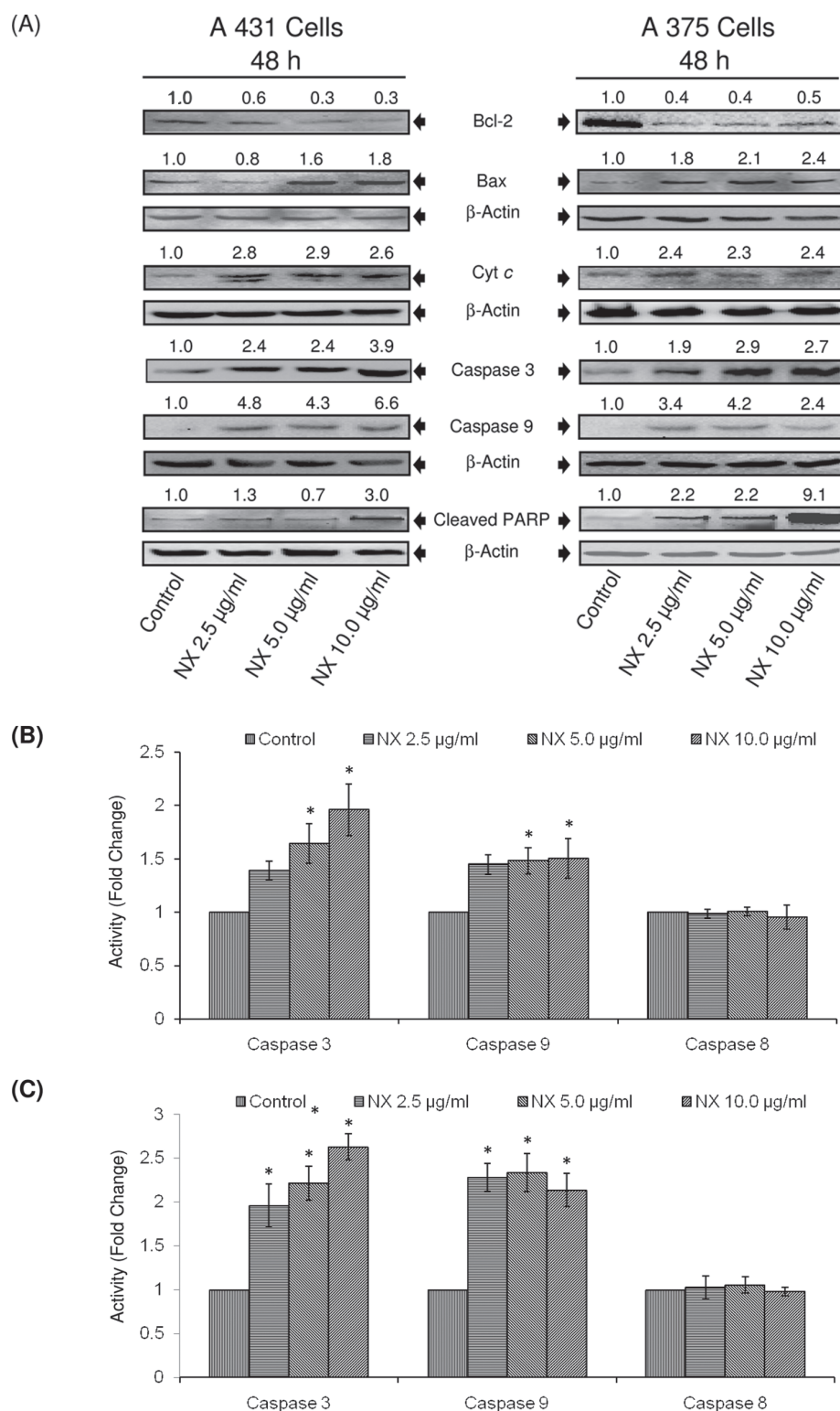


Fig. 5. Effect of NX treatment of A431 and A375 (A) cells on protein expression of Bax, Bcl-2, cytochrome *c*, cleaved caspase 3, cleaved caspase 9 and cleaved PARP. As detailed in text, the cells were treated with NX (2.5–10.0 µg/ml) for 48 h and then harvested. Total cell lysates were prepared and proteins (40 µg) were subjected to sodium dodecyl sulfate–polyacrylamide gel electrophoresis followed by western blot analysis and chemiluminescence detection. Equal loading of protein was confirmed by stripping the blot and reprobing it for β-actin. The blots shown here are representative of three independent experiments with similar results. The values above the figures represent relative density in terms of fold change compared with control after normalization with β-actin. Effect of NX on catalytic activity of caspases 3, 9 and 8 in A431 (B) and A375 (C) cells. As detailed in text, the cells were treated with NX (2.5–10.0 µg/ml) for 48 h and then harvested. Cell lysates were prepared and proteins (100 µg) from each sample were subjected to respective caspase assay using specific substrates conjugated with 7-amino-4-trifluoromethylcoumarin as fluorophore followed by incubation at 37°C for 1 h. The data are shown as fold change compared with control and each value represents the mean ± SE of five experiments in which each treatment was performed in multiple flasks (**P* < 0.05 versus control).

which resulted in inhibition of NF- κ B. Thus, the observed inhibition of NF- κ B by NX could be an appropriate molecular target for cancer prevention, as reported earlier by other investigators (8,30,31). Furthermore, our results also suggest that topical application of NX prior to TPA blocked the phosphorylation and subsequent activation of TPA-mediated MAPKs including ERK1/2, p38 and JNK. In this regard, studies have demonstrated the role of ERK1/2, p38 and JNK in skin carcinogenesis and that chemopreventive agents are capable of preventing skin tumorigenesis by inhibiting MAPK activation (20,32–34). As revealed in our short-term studies, NX showed substantial inhibition against TPA-induced inflammatory parameters and cell proliferation in mouse skin, which indicate that NX may have modulating potential at the promotional step of skin tumorigenesis. Furthermore, topical application of NX prior to TPA application in DMBA-initiated mouse skin showed substantial reduction of cumulative number of tumors, percentage of mice with tumors, number of tumors per tumor-bearing mouse and number of tumors per mouse. These data suggest the anti-skin tumor-promoting potential of NX and the chemoprevention is due to its antiproliferative activity.

Earlier reports with prostate cancer cells showed that NX also has chemotherapeutic potential—it arrests cell cycle progression at the G₀/G₁ phase; activates Bax and inhibits Bcl-2; and induces apoptosis (7). Therefore, in an attempt to assess the chemotherapeutic effects of NX against skin cancer, we chose two most common skin cancer cells, human squamous carcinoma A431 and human melanoma A375 and examined the cytotoxic potential of NX against these cells. Our results indicate that NX treatment to A431 and A375 cells resulted in significant growth inhibition and apoptotic cell death in these cells. Apoptotic cell death represents a universal and exquisitely efficient suicidal pathway and an ideal way for elimination of unwanted cells; however, cancerous cells show dysregulation of this mechanism, which makes the cell virtually immortal and resistant to stress stimuli as well as therapeutic agents (35). Therefore, the apoptotic pathway is widely studied as a potential target for cancer chemotherapy (36,37). It is well documented that apart from deregulation of apoptotic mechanism, alteration in cell cycle regulation may also result in unwanted cellular proliferation, which ultimately leads to the development of cancer (38,39). The present study showed that NX treatment of A431 and A375 cells resulted in a dose-dependent arrest of cell cycle progression at the G₀/G₁ phase and thus cell cycle regulation-mediated apoptosis may be considered an important mechanism of NX-mediated cell growth inhibition. In this regard, studies have shown that various chemotherapeutic phytochemicals possess the ability to induce apoptosis in cancer cells by arresting cell cycle progression in various phases of cell division (37,40,41). Furthermore, NX treatment to A431 and A375 cells results in significant decrease in the levels of Bcl-2 protein along with an increase in the levels of Bax protein, thus enhancing the Bax/Bcl-2 ratio, which favors apoptosis. The observed increase in Bax/Bcl-2 ratio acts as a proapoptotic signal, resulting in the release of cytochrome *c* from mitochondria to cytoplasm (42,43). The NX-induced cytochrome *c* release in the cytoplasm activates the apoptosome, which leads to auto-activation of caspase 9; this further cleaves procaspase 3 to its activated form caspase 3, the executioner caspase (44). Caspases are the mediators of the execution mechanism of apoptosis, and their activation results in the cleavage of the PARP protein—a DNA repair enzyme in the cell—and subsequent DNA degradation and apoptotic death (18). Because caspase 8 was not found to be activated after NX treatment in both the cells, it can be deduced that NX-induced apoptosis is mediated via activation of the intrinsic pathway. Interestingly, NX-induced loss of viability was found to be substantially more in human melanoma A375 cells compared with human squamous carcinoma A431 cells, suggesting that NX may be useful in the treatment of the most aggressive form of skin cancer. Moreover, due to minimal or no cytotoxic effect of NX (2.5–10.0 μ g/ml) to HaCaT and PMKs but significant reduction in cell viability and apoptotic cell death in A431 and A375 cells, it can be deduced that NX may have high therapeutic index for skin cancer.

In summary, the results of the present study indicate that NX may be a potential candidate for prevention/therapeutic strategies against

skin and other epithelial cancers in humans. A positive outcome of such studies could specifically be beneficial as NX is already in clinical use for pain management and is consumed as dietary supplement around the world including USA.

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